

a reel for flexibility and can be cut to suit any axis length from 10mm up to 50m and beyond on special orders, ensuring zero waste. Inside the read head, LED emits light onto the angled scale graduations, where it is directed back through a transparent phase grating, producing sinusoidal interference fringes at the detection plane. Signals not matching the scale period are filtered out to maintain signal stability.

The lightweight scale ensures negligible loading on the voice coil motor and can be mounted onto the moving part directly using adhesive backing tape without any extra weight from bolts and clamps. Easy and fast installation ensures a cost-effective assembly. For further stability and lower sub-divisional error, its analogue version can be connected to REE

interface, which includes automatic offset control (AOC), automatic gain control (AGC) and automatic balance control (ABC) for efficient operation.

In another instance, a laser-marking systems supplier chose MicroE's PCB-mounted Mercury 1200 encoder (15.3mm x 12.7mm x 5.6mm) for its servo control system with incremental sine/cosine feedback. Along with space and cost savings, it provides an analogue output that can be interpolated for resolution up to 4.2 million counts per revolution. The encoder weighs only two grams and consumes low power, which helps in maintaining thermal stability of the system. Wide alignment tolerance ensures ease of installation. Besides, it can achieve up to 120nm cyclical error. Coupled with a 12mm diameter opti-

cal disk, the encoder provides 200,000 counts per revolution after seven-bit interpolation in client's electronics, according to azosensors.com.

High precision in movement is important in other similar applications ranging from office equipment such as photocopiers to medical equipment. In addition to saving time, laser technology improves safety and results by utilising miniature optical encoders with high resolution and accuracy to focus exactly on the target spot in surgeries usually done on the human eye, giving it an edge over traditional methods. Care must be taken for protecting them in case of harsh environments. With recent advancements and growing market presence, use of encoders is bound to increase in the future.

—Ayushee Sharma

3 SMART TICKETING EASING THE WAY WE TRAVEL

Smart ticketing is changing the way we travel, providing greater convenience and security to travellers and transport operators

As technology evolves, public transport will soon be a whole lot easier to use than it is today. In earlier days, if you forgot your ticket somewhere, or it was stolen, you were not able to prove you had bought it. Paper ticketing also suffers from mishandling, as it could get washed away or torn out. But now, with smart ticketing you can have your e-ticket on email and mobile. This gives you the freedom to access your ticket anywhere, anytime.

Smart ticketing is changing the way we travel, providing greater convenience and security to travellers and transport operators. Use of smart cards and smartphones enables travellers to store and use their tickets electronically. This eliminates the need for traditional paper tickets and enables users to simply tap their smart card or phone to access their transport.

The combination of back-office processing and analytics systems with Web portals, mobile applications and smart cards delivers comprehensive solutions to transport operators. Data

analytics enable optimisation of smart transport schemes for higher profit through access to real-world travel data. This helps in easy management of transaction data to ensure accurate reimbursements.

Smart tickets broadly fall into two categories: barcode (or QR code) and smartcard. Barcode tickets have a barcode that allows devices to optically read and interpret content of the tickets, whereas smartcard tickets store the tickets on a plastic card and allow contactless readers to read the content using a radio.

Both physical and virtual smart cards offer secure and easy-to-implement solutions for any transport scenario, including rail, bus, air or metro. Ticketing apps provide travellers with an intuitive interface to purchase, manage and use tickets on a mobile device. These allows customers to download tickets quickly and securely over the Internet. The host operator or processing system manages security of transport operator's devices and processes the transactions coming

from those devices. Card management system manages customer information and cards to enable operators to issue new cards and provide customer support.

Smart tickets help reduce overcrowding at busy stations and provide customers with more choice, time and freedom. The aim is to provide customers with tailored information and services so that travelling by rail, aircraft, bus or metro becomes a seamless part of an end-to-end journey.

Smart tickets enable operators to reduce cost, generate new revenues and innovate services with high reliability, security and memory capabilities. Confidex contactless smart tickets are one such example, which have been specially designed for large-scale ticketing applications in smart cities.

Smart cards with RFID tags allow users to manage assets in harsh outdoor environments. This also helps them track, identify and authenticate indoor assets like metals in the manufacturing industry. Enhanced security and better performance features enable automated